

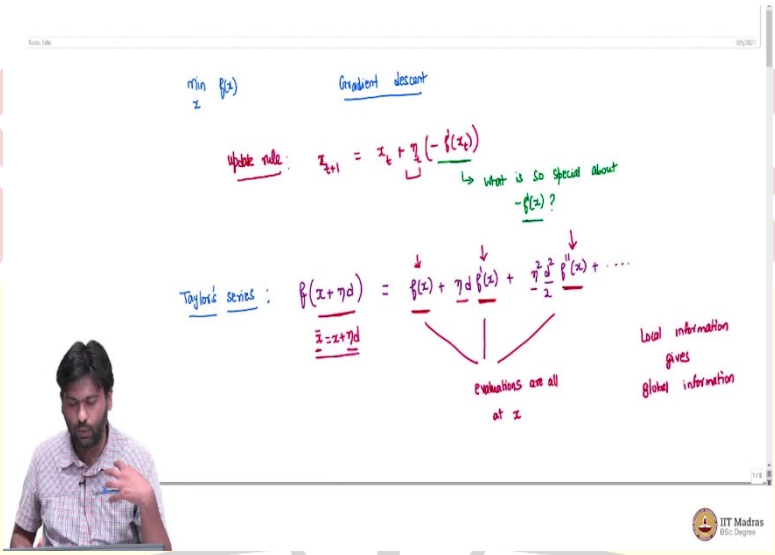
IIT Madras

ONLINE DEGREE

MLF
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Gradient Descent and Taylor Series

Hello, everyone, welcome back, we will continue our discussion on optimization and last time, we saw this fundamental or basic algorithm and optimization for unconstrained problems.

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Where you were trying to minimize a function, f(x)without any constraint on x. So, and we gave this algorithm called Gradient Descent, though we did not describe what a gradient is, but this is the algorithm that we described and one of the key equations or the key update rule of this algorithm was as follows.

At any point x t, you want to update your current point to x t + 1 because gradient descent is an iterative algorithm, the way we were doing it as follows x t + 1 equals x t + sun η t × - f'(x)t. So, this was the simple update rule. Basically, you take a gradient step or a derivative step, but then in the negative direction and you do not take a 1-unit step, but then you take a step size of η and we saw that this algorithm would actually do well, in the sense that it will find the direction for some simple problems.

And I kind of told that this is true in general, that this is a good way to decide on the direction, but I did not, justify that. So, now today, what we are going to do is we are going to ask the question, what is so special about - f'(x)t, what is so special about this? About - f ' of x.

So, why would one want to intuitively move in the direction of the negative derivative. So, what is the importance of this? So, this is the first question that we are going to ask today and

there are different ways to answer this and the way we will do it is as follows. So, I am going to remind you of a popular equation, which is called as the Taylor series and I will describe this and write the equation down and then explain what it means.

So, here is the equation, it might look a bit unwieldy, but I will try to explain what it actually means. So, you want to compute the function, some functions value at some point, let us say x + η × d and we will see why I am writing it this way and Taylor series says that the functions value at a particular point can be expanded as follows, f(x)+ η × d × f'(x)+ η squared d squared by 2 f " of x + . . . and so on.

So, what are we saying here? Of course, this is a, this looks like a complicated way to expand functions and so on. But, what is the main fundamental reason, fundamental importance of this Taylor series. So, the importance is as follows and the importance specifically for an optimization point of view will become apparent in a minute, but in general the importance is that this is a very beautiful equation which says that, let us say I want to understand the functions value or functions behaviour, which means the functions value at some other point.

So, at some point x + η t, so we could simply call this some other point x ', maybe I would not use x ', so maybe some other point x hat which is x + η d and we want to understand what is the value of this function at x bar, which is x + η d and now Taylor series says that the functions value is equal to an infinite expansion, but what is this right hand side look like.

So, if we look at the hand side and see where is the function being used in the right-hand side, you will notice that it is being used here, here, here and so on. So, in each of these cases, you are looking at the functions value f(x), you are looking at the functions derivative at x f'(x)second derivative at x f "(x) and so on.

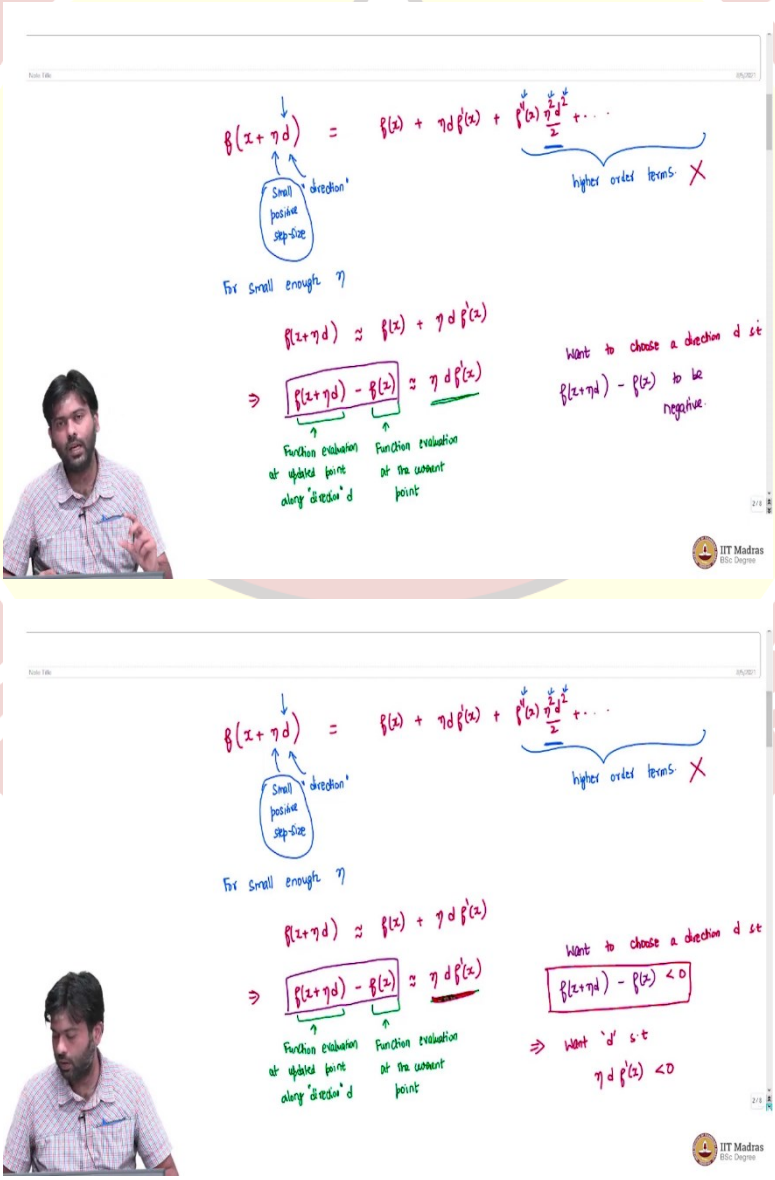
But then the key point to note here is that the evaluations of the function or its derivatives are all at x. So, which means that no matter which point x bar, where I want to know how the functions value looks like I can get that value if I know everything about the function at x, by everything I mean that if you know the functions value at x, if you know the functions derivative at x, if you know the function double derivative of x and so, on.

If you know complete information locally. So, then you can compute the functions value at any point and this is amazing in the sense that the functions complete description is present locally. So, if you know what is the slope, if you know what is the curvature, which is the second derivative and so on at a current point, then you can compute the function at any other point.

So, of course, it has to be scaled by η square η d and so on some function of η and d will come × picture, because depending on which point you want to understand what the function is doing. So, in some sense, local information gives global information, so that is the fundamental idea of Taylor series.

Which means that if I know the functions behaviour completely at a given point x, then I know the function behaviour at every other point. So, because you can always set η and d accordingly to get any x bar, so that you can evaluate the function at any other point. Now, for our purposes, for the purposes of optimization, why is this useful? Well, let us see why this might be useful for optimization.

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So, what are we trying to do, we are trying to understand the functions value at x + η × d. Now, what is so special about this form x + η × t. Well, we have seen in our gradient descent algorithm, you can think of η as a small positive step size and d is some direction in which you want to move.

So, you want to move in the direction of d, but then you are only taking a step size of η. Now, after I make the step size, so I am at a current point x and then I am, I do not know which d I want to pick. So, let us say I fix some d and then I move η towards in the direction of d, then the functions value at the point that I will end up in will be f(x)+ η d.

So, remember the gradient update rule, which was something like x t + 1 was x t - η × the derivative. We will come to that in a minute. But so instead of the derivative, let us say there is some general direction d in which I am trying to move and I am trying to ask what is the function value at this, after they take the step?

Now, we know from Taylor series that of course, this is going to depend on f(x)+ η × d f'(x)and f " of x η × η squared d squared by 2 + and so on. So, this is an exact equality, that is what I wrote earlier. Now, because η is a small positive step size, sorry about that.

Because it is a small positive step size, think of this as something like let us say 0.1, 0 naught 1. So, then what can we say about these values. So, if η is very small, then η squared is going to be even smaller and then the third term here is going to be η depending on η cube, which is going to be much more smaller.

So, which means that for small enough η everything after the first term, we can think of these as higher order terms. When you say order, it means that the order of the derivative and also the power of the, power by which you raise it. So, these are called as higher order terms and they are going to be really small. So, I can kind of, for very, very small eater to intuitively understand this I can kind of neglect these terms and then say that for small enough η.

So, for small enough η, what happens I can argue that, well f(x)+ η d is going to be something like f(x)+ η × the f ' of x. I am kind of neglecting all these terms. So, they are going to be really, really small. So, which means that I have not put an equal to here, I have just made it approximately equal to, it will be very close to this.

Now, why is this useful? Well, now let us look at the, this implies what? So, this implies the following, x + η d - f(x), I am just bringing the f(x)to the other side. So, this is approximately

it looks like $\eta \times d \times f'$ of x. So, what has happened now, I was in a current step x and what I have is f(x), which is the function evaluation at the current point, the current point x.

So, that is what f(x)itself is. Now, I decide on some direction, I have not fixed what this direction is. But let us say I pick some direction d and then I take η step in that direction, I end up here and now this is the function evaluation at the new point, at the updated point, like along direction d.

So, I am taking an η step along the direction d and then I am asking what is the function value and now this difference can be, it is arbitrarily close to either $\times d \times f'$ of x, that is what your Taylor series says, after removing the higher order terms. Now, why is this useful? Now, we are trying to minimize a function, which means I am at a current point and then I want to go to a different point, where the function value is < the current points value.

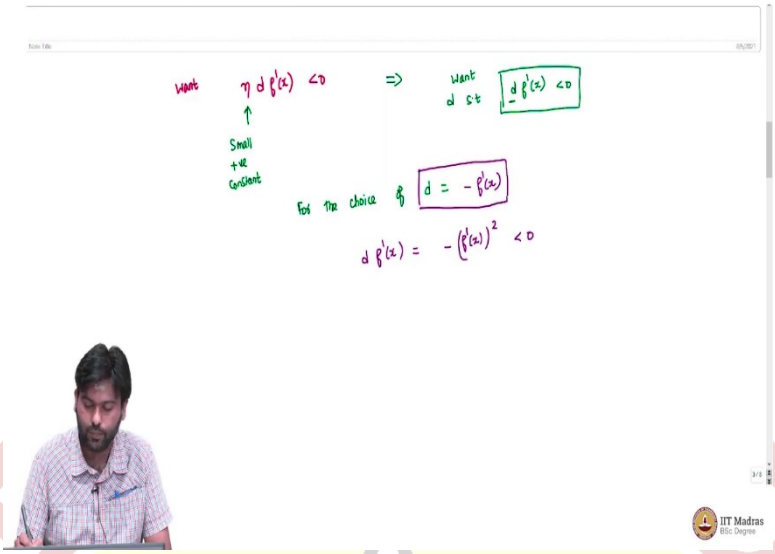
So, then, if I am guaranteed this will happen, then I am kind of moving in the descent direction, so that my functions value will keep reducing and hopefully I will reach the minimum. So, that is the idea. Now, if that has to happen, then what should we, what should happen to this quantity?

If my function value actually ends up decreasing, after I take η step in the direction of t, then I want $f(x)+ \eta$ d - f(x)to be negative. Why do I want it to be negative? Well, if it is negative, then it means that the function value has actually decreased after moving to $x + \eta$ d.

But then this may or may not happen for all directions. So, I mean we do not know if it will happen for all directions, but then we have the power to choose this direction. So, which means that we want to choose a direction d such that $f(x)+ \eta$ d - f(x)is < 0. So, instead of to be negative, I will just say this is going to be, we want this to be an equation < 0.

We have not yet chosen a d. But then if we want a direction to do what we really want to do, which means that minimize the function as we go along the direction, then we want this direction to actually satisfy $f(x)\eta$ d - f(x)to be < 0. But now Taylor series will say that this quantity $f(x)+ \eta$ d - f(x)is actually well approximated by η a $\times d \times f'$ of x. So, which implies want d such that $\eta \times$ d in the f'(x)is < 0, we still have not picked the d, whatever d we pick, we better pick it such that $\eta \times d$ f'(x)is < 0. That is what our Taylor series expansion is saying. So, now how does that help us.

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So, let me read write that again, $\eta \times t \times f'(x)$is < 0, this is want this to happen. Now remember, η is a small positive constant, so we do not have much, I mean η does not change the sign of this, this term, so whatever η you put, it is going to be a positive, a positive scalar. Which means that if $\eta \times d \times f'(x)$is < 0 equivalently, we can say that we want $d \times f'(x)$to be < 0.

So, want to be such that $d \times f(x)$is < 0. So, our Taylor series would tell us that, as long as you are taking a very small step, you pick a direction, such that $d \times f'(x)$is < 0, then the point where you will end up in will have a lesser function value than where you started.

So, which means that we somehow have to pick a d, which is which satisfies $d \times f'(x)$is < 0. Now, f'(x)is just the slope of a function at a particular point, so it is like a single dimensional function, so it is just capturing the slope, the derivative. Now, this derivative could be either positive or negative.

So, which means that a single d cannot work for all possible points of x. So, because f'(x)could be positive or f(x)could be negative, which means that the d that we choose will necessarily depend on the x somehow and now for the choice, now what could be a choice of d?

So, what is a good choice of d which will guarantee that $d \times f'(x)$is < 0? Well, for the choice of d, which we have taken in our gradient descent algorithm as - f ' of x, $d \times f'(x)$, simply becomes - f'(x)square. So, - f'(x)× f ' x - f'(x)squared and we know for sure that this is < 0.

Unless f'(x) is 0. Well, if f ' (x) is 0, we know that we have stopped there anyway. So, now, this is the reason why you can choose t as - f ' x, you can potentially move in the gradients, the derivative negative direction and Taylor series would tell us that as long as your step size is

small, then you are expected your function value is expected to reduce. So, that is the rationale for choosing d as f ‘(x).